Name: __________________________ Roll No.________________
Section: _____________

Computer Networks Final Exam Fall 2012 Total marks: 50 Time allowed: 2.5 hours

NOTE: Write **only** in the space provided for the answers. No extra sheet will be provided

This exam consists of 8 questions on 5 pages (You can use back side of each page for rough work).

Q1. How do you compare Datagram Subnets to Virtual Circuit Subnets? State three differences (6)

Q2. How simultaneous data and voice communication is possible in Digital Subscriber's Line using DSL modem. Why this cannot be done using an ordinary modem? (4+3)

Q3. The polynomial generator $x^4+x^3+x^2+1$ is used for communication between two machines **A** and **B**. Machine **A** wants to send bit stream of 101011111 to machine **B** using the Cyclic Redundancy Check method. Show the actual bit stream transmitted. (4)

Q4. State four advantages of subnetting. Find the maximum number of hosts a network can handle with a subnet mask of 255.255.240.0. (4+2)

Q5. What are Hidden Station and Exposed station problems? Which of these problems is solved using MACA protocol and how? Explain with the help of diagram (4+4)

Q6. What potential problem can occur in distance vector routing protocol. Describe with the help of an example subnet. Do you think this protocol is scalable? Please describe the reason (Scalable means that a protocol performance does not degrade as the network size increases) (5+4)

Q7. How Tomlinson’s solution solves problem of delayed duplicates in data TPDU? (6)

Q8. How three way handshake protocol is used for identifying delayed duplicates in connection TDPUs? (4)

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Computer Networks - Subjective

Total Points: 60 Time Allowed: 2 hours 20 minutes

Instructions:

- **Subjective exam should only be given to students once they return the Objective exam**
- **Extra sheet may be given for rough work, however, it must NOT be collected**

Question1: (2+2+2+4+6)

What should be the sampling rate for reconstructing a filtered signal of bandwidth B?

Answer:

Using Nyquist theorem, which signal property can be used to achieve higher data rates on a channel with bandwidth B?

Answer:

Is there any channel property that Nyquist overlooked while deriving equation for Maximum data rate of a channel? If yes, state property name. If No then state reason why Nyquist theorem is perfect for data rate calculations?

Answer:

Why 35kbps is the maximum Shannon limit for the data rate of the telephone systems (PSTN)?

Signal to noise ratio in a typical telephone system is 30db and the channel bandwidth is 4kHz.

a. What is the maximum bit rate possible?

b. What is the number of voltage levels to achieve this bit rate?

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Question2: (2+3+3)

Following character encoding is used in a data link protocol

A: 01010111

B: 11101011

ESC: 11100111

FLAG: 11101111

Show the bit sequence transmitted for the following data with each of the following framing schemes

Data: A ESC FLAG ESC FLAG B

1. Character Count

---

2. FLAG Bytes with byte stuffing

---

3. Starting and ending flag bytes with bit stuffing (hint: the flag mentioned here is different than the data FLAG given above, remember bit stuffing flag?)

Question3: (4)

The generator polynomial **11101** is agreed between sender and receiver. Assume that the receiver receives the bits **1101010100110**. Is the data received same as the data sent? Show your working.

Answer: _______________

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Question 4: (3+3)

Fill in the following table by writing names of two network devices (devices that are used for inter connections) that are used at each layer

Layer	Device1	Device2
Physical Layer		
Data Link Layer		
Network Layer		

Give one line description of each of these devices. Your description should explicitly state the use of the device at a particular layer of the OSI reference model

Bridge or Hub:____

Router:____
__

Switch:
____

Question 5: (1+1+2)

Suppose there is a 10 Mbps microwave link between a geostationary satellite and its base station on Earth. Every minute the satellite takes a digital photo and sends it to the base station. Assume a propagation speed of 2.4 x 10^8 meters/sec. (remember that Satellites are 36000 km away from earth)

a. What is the propagation delay of the link?

____

b. What is the bandwidth-delay product**,** $\boldsymbol{R \cdot d_{prop}}$?

____

c. Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting?

____

____

____

____

Question 6: (4+6)

a. Do you think the distance vector routing protocol is scalable? Please describe the reason to support your answer (Scalable means that a protocol performance does not degrade as the network size increases).

Your Answer: _______________

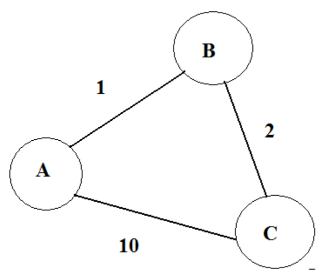
Reason:

____

____

____

b. Calculate the shortest paths using DV algorithm for the following 3 node network. Show all the tables and your working through iterations.



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Question 7: (2+2+1+2+3+2) Consider the figure given below. Assuming TCP Reno is the protocol experiencing the behavior shown below, answer the following questions

a. Identify the intervals of time when TCP congestion avoidance is operating. Explain.

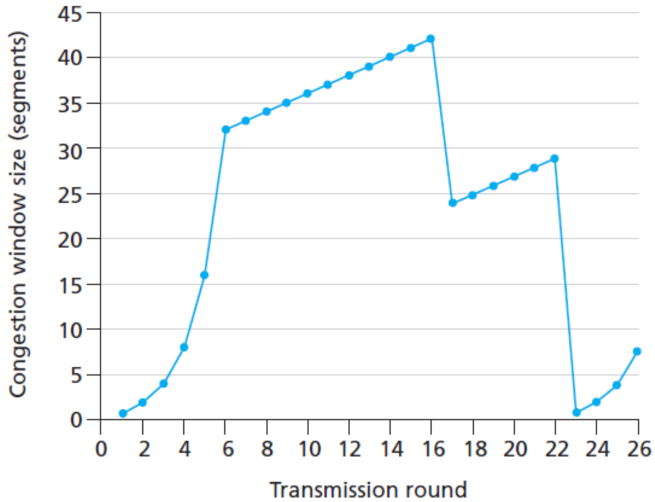
b. After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout? Explain

c. What is the initial value of ssthresh at the first transmission round?

d. What is the value of ssthresh at the 18th transmission round? Why?

e. Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh? Explain

f. Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 17th round?


Congestion window size (segments)
45
40
35
30
25
20
15
10
5
0
0 2 4 6 8 10 12 14 16 18 20 22 24 26
Transmission round

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Course Name:	Computer Networks	Course Code:	CS307
Program:	BS(CS)	Semester:	Fall 2018
Duration:	3 hours	Total Marks:	50
Paper Date:	21-12-2018	Weight	50
Section:	ALL	Page(s):	10
Exam Type:	Final		

Student : Name:_________________________ **Roll No.**______________ **Section:**______

Instruction/Notes: Attempt questions on this paper. You may use rough sheet but it should not be attached to this paper as it will not be marked.

1. Multiple choice questions. Please write your answer/section in the table given below. Answers outside the table will not be considered. [10 Marks]

1.1		1.6	
1.2		1.7	
1.3		1.8	
1.4		1.9	
1.5		1.10	

1.1 ICMP is primarily used for
a) error and diagnostic functions
b) addressing
c) forwarding
d) none of the mentioned

1.2 User datagram protocol is called connectionless because
a) all UDP packets are treated independently by transport layer
b) it sends data as a stream of related packets
c) it is received in the same order as sent order
d) none of the mentioned

1.3 An endpoint of an inter-process communication flow across a computer network is called
a) socket
b) pipe
c) port
d) none of the mentioned

1.4 Transport layer protocols deals with
a) application to application communication
b) process to process communication
c) node to node communication
d) none of the mentioned

1.5 For a 10Mbps Ethernet link, if the length of the packet is 32bits, the transmission delay is (in milliseconds)
a) 3.2
b) 32
c) 0.32
d) None of the mentioned

1.6 The time required to examine the packet's header and determine where to direct the packet is part of
a) Processing delay
b) Queuing delay
c) Transmission delay
d) All of the mentioned

1.7 The computation of the shortest path in OSPF is usually done by
a) Bellman-ford algorithm
b) Routing information protocol
c) Dijkstra's algorithm
d) Distance vector routing

1.8 Correct order of the operations of OSPF
1. Hello packets
2. Propagation of link-state information and building of routing tables
3. Establishing adjacencies and synchronisation database

a) 1-2-3
b) 1-3-2
c) 3-2-1
d) 2-1-3

1.9 Which of this is not a class of IP address?
a) ClassE
b) ClassC
c) ClassD
d) ClassF

1.10 The TTL field in an IP datagram/packet has a value of 10. How many routers (max) can process this datagram?
a) 11
b) 5
c) 10
d) 1

2. Consider the network scenario given in Figure 1. Assume that we know the bottleneck link along the path from the server to the client is the first link with rate $\mathbf{R_s}$ **bits/sec**. Suppose we send a pair of packets back to back from the server to the client, where client is connected to the router with $\mathbf{R_c}$ **bits/sec** and there is no other traffic on this path. Assume each packet of size **L bits**, and both links have the same propagation delay given by $\mathbf{d_{prop}}$. **[1+2+2=5 Marks]**

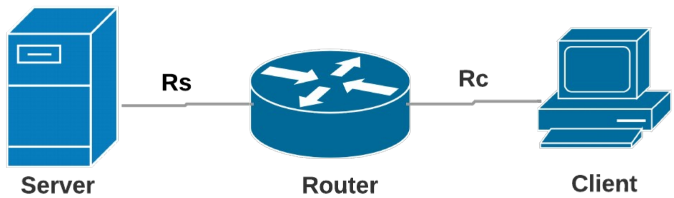

Figure. 1

2.1 What is the packet inter-arrival time at the destination (client)? Meaning how much time elapses from when the last bit of the first packet arrives until the last bit of the second packet arrives?

2.2 Now assume a change in the above scenario where instead of first the second link is the bottleneck (i.e. $R_c < R_s$). Would it be possible that the second packet gets queued at the input queue of the second link? Explain.

2.3 Now suppose that the server sends the second packet T seconds after sending the first packet. How large must T be to ensure no queuing before the second link? Explain. [Use the same bottleneck as 2.2 i.e. $R_c < R_s$.]

3. Suppose the current TCP window size is 28000 bytes and the maximum segment (packet) size is 1400 bytes. The congestion window is maintained in bytes. In slow start, each incoming new ACK increments the congestion window by the maximum segment size. Suppose the sender previously received cumulative ACK 135801 (i.e., the receiver says it received all bytes sent to it, up to and including, sequence number 135800). The sender next receives cumulative ACK 140001. Assuming the sender is in slow start and has enough data in its transmit socket buffer, calculate the number of bytes it now sends. Show all of your calculations clearly. **[5 Marks]**

Bytes the sender can send: _______________

Calculations:

4. Assume a general network scenario comprising a client, a server and three routers between them. Ignoring fragmentation, an IP datagram is sent from the client to the server. Answer the following short questions. **[1+2+2=5 Marks]**

4.1 How many interfaces in total will this IP datagram traverse (i.e. the datagram will travel through how many interfaces)?

4.2 How many forwarding tables will be indexed to successfully move the datagram from the client to the server?

4.3 Suppose TCP segments are being encapsulated in the IP datagrams being sent between the client and the server. When the server receives the datagram, how does the network layer on the server side know which upper layer protocol it should pass the segment to. How does the server-side determine that the payload of the datagram should be given to TCP rather than UDP or (something else)?

5. Consider the network scenario given in Figure 2 comprising four end-systems/hosts: A, B, C and D with MAC addresses: MAC-A, MAC-B, MAC-C, MAC-D and IP addresses: IP-A, IP-B, IP-C, IP-D, respectively. DHCP is enabled for the given network topology. S1 and S3 denote the switches while routers are represented by R1 and R3. Connecting lines depict the network connections (wired) between all the network nodes with individual digits on each end representing the interface number of the connected node.
Furthermore, for IP and MAC addresses of the interfaces, IP-(Interface number) and MAC-(Interface number) format is in use. For example, IP-9 and MAC-9 refers to the IP and MAC addresses of Router1's interface 9.

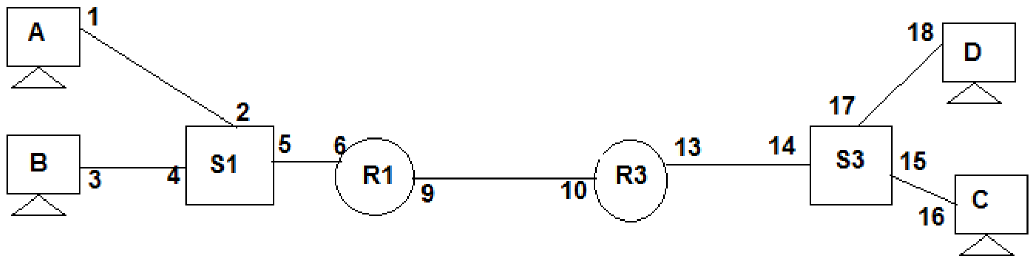

Figure. 2

Consider the following ARP table for interface 6 of the router R1 and specify in the entries given are correct or not. For each entry, please give a brief reason for your choice. [5 Marks]

IP Address	MAC Address	Correct Entry? (Yes/No)	Brief Reason
IP-A	**MAC-A**		
IP-D	**MAC-D**		
IP-9	**MAC-9**		
IP-5	**MAC-5**		
IP-3	**MAC-B**		

6. IP Subnetting. Please use the space provided to show your working. [2.5+2.5=5 Marks]

6.1 Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.15.17/24. Also suppose that Subnet 1 is required to support at least 58 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 15 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.

Subnet address 1: ____

Subnet address 2: ____

Subnet address 3: ____

Rough work (6.1):

6.2 For the following IP address and subnet mask **10.x.y.z /30**. List down all the network prefixes possible, listing the difference/spacing between each subnetwork. [Hint: calculate all the available values/range of x, y and z for subnet: /30]

Range of second octet (x): __________________ spacing between successive values: ____________

Range of third octet (y): __________________ spacing between successive values: ____________

Range of fourth octet (z): __________________ spacing between successive values: ____________

Rough work (6.2):

7. Suppose that TCP's current estimated values for the round trip time (estimatedRTT) and deviation in the RTT (DevRTT) are 210 msec and 49 msec, respectively. Suppose that the next three measured values of the RTT are 280, 210, and 220 respectively.

Compute TCP's new value of estimatedRTT, DevRTT, and the TCP timeout value after each of these three measured RTT values is obtained.Use the values of $\alpha = 0.125$ and $\beta = 0.25$. **[1+2+2=5 Marks]**

After the first RTT estimate is made:

estimatedRTT =
DevRTT =
TimeoutInterval =

After the second RTT estimate is made:

EestimatedRTT =

DevRTT =

TimeoutInterval =

After the third RTT estimate is made:

EestimatedRTT =

DevRTT =

TimeoutInterval =

Rough work (7):

8. Consider the Figure 3 below, which plots the evolution of TCP's congestion window at the beginning of each time unit (where the unit of time is equal to the RTT); TCP sends a "flight" of packets of size cwnd at the beginning of each time unit. The result of sending that flight of packets is that either *(i)* all packets are ACKed at the end of the time unit, *(ii)* there is a timeout for the first packet, or *(iii)* there is a triple duplicate ACK for the first packet. In this problem, you are asked to reconstruct the sequence of events (ACKs, losses) that resulted in the evolution of TCP's cwnd shown below. **[5 Marks]**

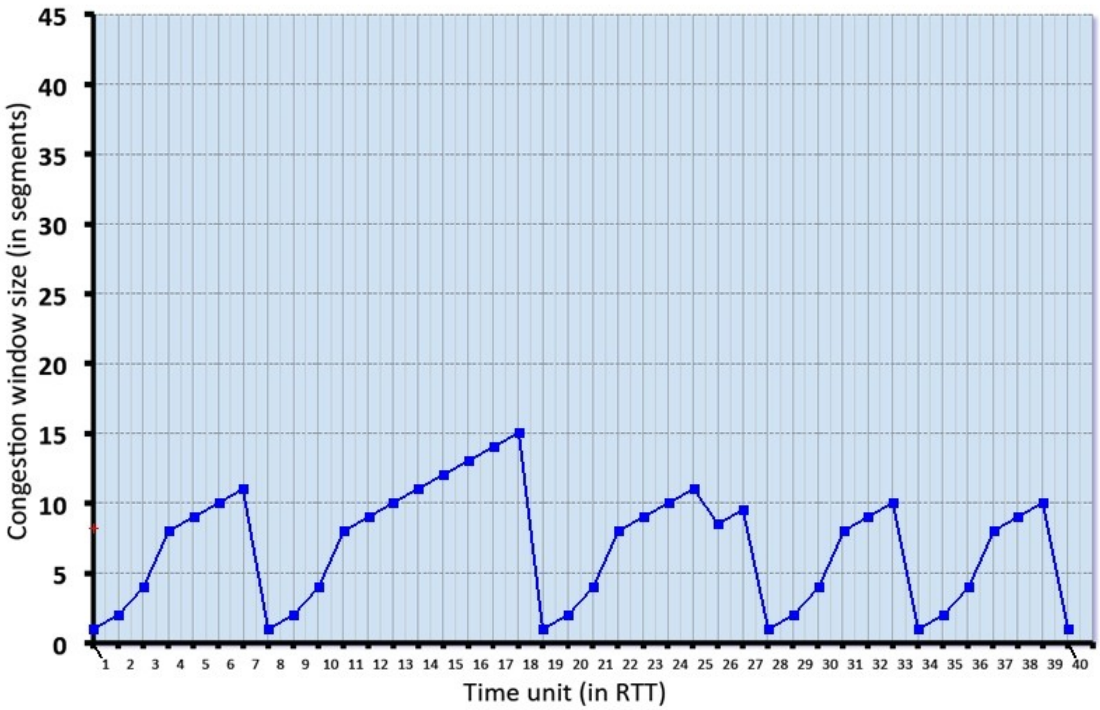

Figure. 3

The initial value of cwnd is 1 and the initial value of ssthresh is 8. Now answer the following questions:

8.1 Give the times at which TCP is in slow start and fast recovery at the start of a time slot, when the flight of packets is sent.

Slow-start: *format [a,b] [x,y] [z]*:
Fast recovery:

8.2 Give the times at which the first packet in the sent flight of packets is lost, and indicate whether that packet loss is detected via timeout, ot by triple duplicate ACKs.

Timeout Loss Interval Points:
3 Duplicate ACK Loss Points:

9. Consider the scenario in Figure 4, in which three hosts, with private IP addresses 10.0.1.12, 10.0.1.16, 10.0.1.20 are in a local network behind a NATted router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router's interface on the LAN side has IP address 10.0.1.26, while the router's address on the Internet side has IP address 135.122.202.209. **[5 Marks]**

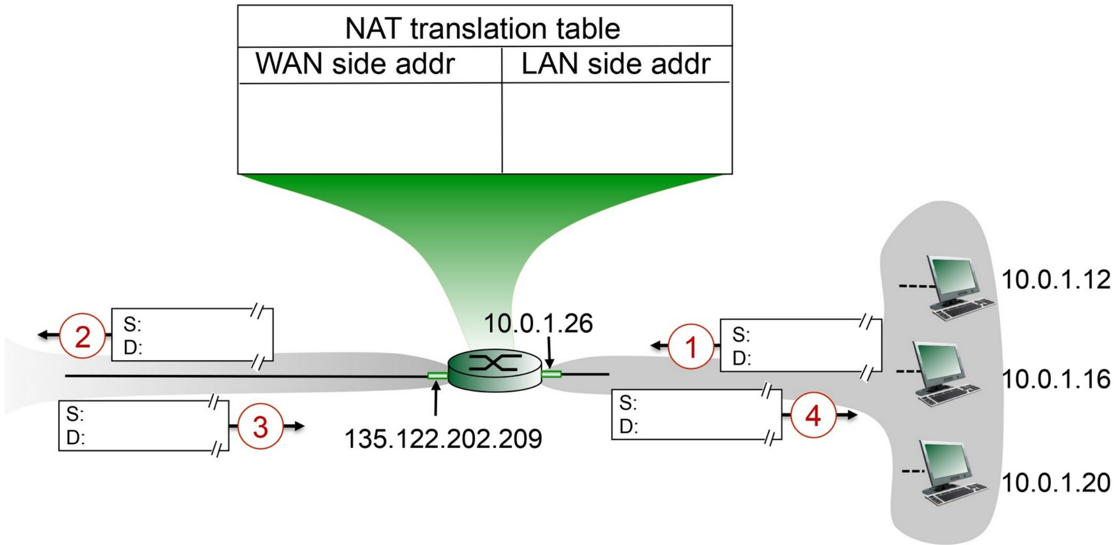

Figure. 4 Network Address Translation Scenario

Suppose that the host with IP address 10.0.1.16 sends an IP datagram destined to host 128.119.166.183. The source port is 3303, and the destination port is 80.

9.1 Consider the datagram at step 1, after it has been sent by the host but before it has reached the NATted router. What are the source and destination IP addresses for this datagram? What are the source and destination port numbers for the TCP segment in this IP datagram?

Source Socket:
Destination Socket:

9.2 Now consider the datagram at step 2, Specify the entry that has been made in the router's NAT table. Make any assumptions as required.

Changes in Source Socket (if any):
Changes in Destination Socket (if any):

9.3 Now consider the datagram at step 3, just before it is received back by the NATted router. What are the source and destination IP addresses for this datagram? What are the source and destination port numbers for the TCP segment in this IP datagram?

Source Socket:
Destination Socket:

9.4 consider the datagram at step 4, after it has been transmitted by the NATted router but before it has been received by the host. What are the source and destination IP address for this datagram? What are the source and destination port numbers for the TCP segment in this IP datagram?

Source Socket:
Destination Socket:

9.5 Identify the differences in datagram's IP addresses and port numbers between step 3 and step 4. Has a new entry been made in the router's NAT table, or removed from the NAT table? Explain your answer.

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Course Name:	Computer Networks	Course Code:	CS307
Degree Program:	BS (CS)	Semester:	Fall 2019
Exam Duration:	150 Minutes	Total Marks:	70
Paper Date:	12-Dec-2019	Weight	40
Section:	ALL	Page(s):	6
Exam Type:	Final Exam		

Student : Name:_______________________________ Roll No.________________ Section:_______

Instruction/Notes:
- Attempt all questions on the provided question paper.
- Even if you use rough sheets, they should NOT be attached with final paper.
- No need to ask questions. If you have confusions, take assumptions where needed.

Question 01: Answer the multiple-choice questions by choosing one option. Fill the provided table with answers. <u>Any answers outside the table will NOT be marked</u>.

1	**C**	**6**	**A**
2	**C**	**7**	**B**
3	**A**	**8**	**D**
4	**D**	**9**	**C**
5	**C**	**10**	**C**

1 Which of the following mapping does Address Resolution Protocol (ARP) provide to the host:
   a. IPv4 to IPv6
   b. Hostname to IP address
   c. IP address to MAC address
   d. MAC address to interface ID

2 Which of the following is NOT an algorithm used to determine the best routing path in computer networks.
   a. Bellman-Ford Algorithm
   b. Dijkstra's Algorithm
   c. Brent's Algorithm
   d. None of the above

3 Which is the following header fields does a router modify while fragmenting an IPv4 packet.
   a. Flag, Identifier, and offset
   b. Header length, offset, and flag
   c. Protocol, header length, and identifier
   d. Destination IP, Source IP, and flag

4 It is possible to detect and correct multiple bit errors on link layer using
   a. CRC
   b. Ethernet
   c. 2-D parity scheme
   d. None of the above

5 Which of the following is NOT a server-side connection state?
   a. SYN_RCVD
   b. ESTABLISHED
   c. TIME_WAIT

d. CLOSE_WAIT

6 Multiplicative decrease in TCP congestion control means
   a. Decrease the congestion window size by half
   b. Decrease the *ssthreshold* value by half
   c. None of the above
   d. Both a & b

7 A TCP connection consists of
   a. 1 socket
   b. 2 sockets
   c. 3 sockets
   d. 4 sockets

8 Which of the following is a stateless protocol?
   a. TCP
   b. GBN (Go-Back-N)
   c. Stop and wait
   d. None of the above

9 The maximum size of an IPv4 header is
   a. 20 bytes
   b. 40 bytes
   c. 60 bytes
   d. minimum 40 bytes

10 Which layer links the network support layers and user support layers
   a. session layer
   b. data link layer
   c. transport layer
   d. network layer

[20 points] Question 2. A certain organization has been assigned a network address block 201.180.128.0/23.

It has been determined that the organization needs:

- 1 network with at least 240 hosts
- 1 network with at least 55 hosts
- 1 network with at least 28 hosts
- 2 networks with at least 15 hosts

a) Design the complete IP addressing scheme for this organization and **fill in the table below**. Show all your work with appropriate comments (if any). **[15 points]**

Table I

Network	**Network Address**	**Subnet mask**	**First available host address**	**Last available host address**	**# of available host addresses**
Network 1	201.180.128.0	/24	201.180.128.1	201.180.128.254	254
Network 2	201.180.129.0	/26	201.180.129.1	201.180.129.62	62
Network 3	201.180.129.64	/27	201.180.129.65	201.180.129.94	30
Network 4	201.180.129.96	/27	201.180.129.97	201.180.129.126	30
Network 5	201.1180.129.128	/27	201.180.129.129	201.180.129.158	30

b) An internet user wants to examine the path that their data packets follow while accessing www.google.com, for which they use *traceroute* network tool. **[5 points]**

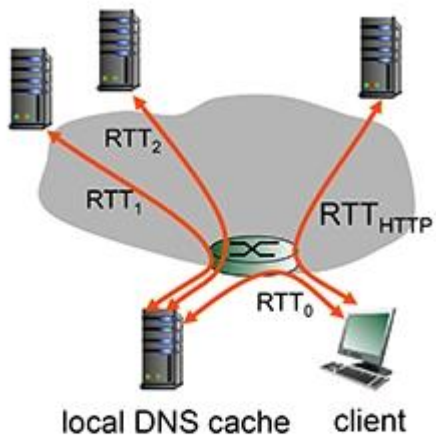
i. What layer-3 *protocol* does *traceroute* use?

ICMP.

ii. Explain (in detail) how *traceroute* uses the available network infrastructure to achieve the desired goal?

3 probes are sent with an increasing TTL value each time to know the number of intermediate nodes between source and destination. This procedure is halted when the probes reach the destination.

[20 points] Question 3: Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your localhost, so a DNS lookup is necessary to obtain the IP address. Suppose that three DNS servers are visited before your host receives the IP address from DNS. The first DNS server visited is the local DNS cache, with an RTT delay of RTT_0 = 3 msecs. The second and third DNS servers contacted have RTTs of 27 and 49 msecs, respectively. Initially, let's suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Suppose the RTT between the local host and the Web server containing the object is RTT_{HTTP} = 37 msecs.



1. Assuming zero transmission time for the HTML object, how much time elapses from when the client clicks on the link until the client receives the object?

2. Now suppose the HTML object references 10 very small objects on the same web server. Neglecting transmission times, how much time elapses from when the client clicks on the link until the base object and all 10 additional objects are received from web server at the client, assuming non-persistent HTTP and no parallel TCP connections?

3. Repeat 2. above but assume that the client is configured to support a maximum of 5 parallel TCP connections, with non-persistent HTTP.

4. Repeat 2. above but assume that the client is configured to support a maximum of 5 parallel TCP connections, with persistent HTTP.

Answers:

1.

The time from when the Web request is made in the browser until the page is displayed in the browser is: RTT0 + RTT1 + RTT2 + 2*RTTHTTP = 3 + 27 + 49 + 2*37 = 153 msecs.

Note that 2 RTTHTTPs are needed to fetch the HTML object - one RTTHTTP to establish the TCP connection, and then one RTTHTTP to perform the HTTP GET/response over that TCP connection.

2.

The time from when the Web request is made in the browser until the page is displayed in the browser is: RTT0 + RTT1 + RTT2 + 2*RTTHTTP + 2*10*RTTHTTP = 3 + 27 + 49 + 2*37 + 2*10*37 = 893 msecs.

Note that two RTTHTTP delays are needed to fetch the base HTML object - one RTTHTTP to establish the TCP connection, and one RTTHTTP to send the HTTP request, and receive the HTTP reply. Then, serially, for each of the 10 embedded objects, a delay of 2*RTTHTTP is needed - one RTTHTTP to establish the TCP connection and then one RTTHTTP to perform the HTTP GET/response over that TCP connection.

3.

The time from when the Web request is made in the browser until the page is displayed in the browser is: RTT0 + RTT1 + 2*RTTHTTP + 2*RTTHTTP + 2*RTTHTTP = 3 + 27 + 49 + 2*37 + 2*37 + 2*37 = 301 msecs.

As in 2. above, two RTTHTTP delays are needed to fetch the base HTML object - one RTTHTTP to establish the TCP connection, and one RTTHTTP to send the HTTP request, and receive the HTTP reply containing the base HTML object. Once the base object is received at the client, the maximum of five requests can proceed in parallel, each retrieving one of the 10 embedded objects. Each (in parallel) requires two RTTHTTP delays - one RTTHTTP to set up the TCP connection, and one RTTHTTP to perform the HTTP GET/response for an embedded object. Once these first five objects have been retrieved, the remaining 5 embedded objects can be retrieved (in parallel). This second round of HTTP GET/response to retreive the remaining 5 embedded objects takes two more RTTHTTP delays.

4.

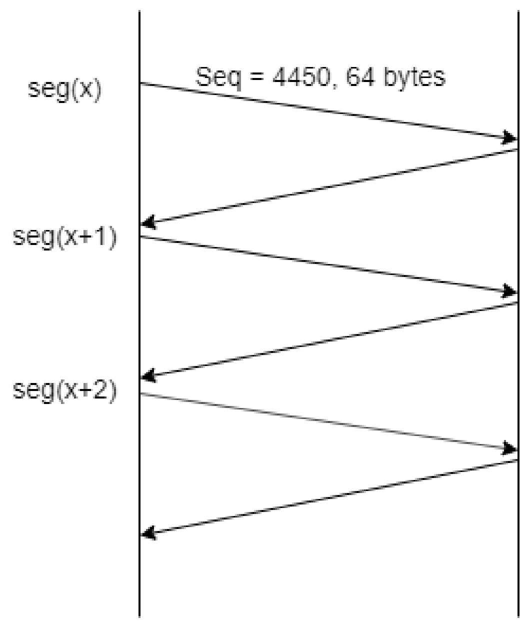
The time from when the Web request is made in the browser until the page is displayed in the browser is: RTT0 + RTT1 + RTT2 + 2*RTTHTTP + RTTHTTP + RTTHTTP = 3 + 27 + 49 + 2*37 + 37 + 37 = 301 msecs.

As in 2. and 3. above, two RTTHTTP delays are needed to fetch the base HTML object - one RTTHTTP to establish the TCP connection, and one RTTHTTP to send the HTTP request, and receive the HTTP reply containing the base HTML object. However, with persistent HTTP, this TCP connection will remain open for future HTTP requests, which will therefore not incur a TCP establishment delay. Once the base object is received at the client, the maximum of five requests can proceed in parallel, each retrieving one of the 10 embedded objects. Each (in parallel) requires only one RTTHTTP delay to perform the HTTP GET/response for an embedded object. Once these first five objects have been retrieved, the remaining 5 embedded objects can be retrieved (in parallel). This second round of HTTP GET/response to retreive the remaining 5 embedded objects takes only one more RTTHTTP delays, since the TCP connection has remained open.

[20 points] Question 4: Suppose Node A (sender) and B (receiver) have a TCP connection between them. Assume that a single segment seg (x-1) is timed out. Consider the size of the TCP receiver buffer is 300 bytes. Assuming all packets of equal size i.e. 64 bytes, if *ssthreshold* = 6, then **answer the following questions in <u>table given below</u>** by looking at the provided figure:

1) Provide
    a. Sequence number of seg (x+2)
    b. Acknowledgement of seg (x+2)
2) Sequence number of bytes of seg (x+4)
3) *Receiver window* field value in acknowledgment of seg (x+3)
4) *Receiver window* field value in acknowledgement of seg (x+4)
5) Value of *window size* and *ssthreshold* after acknowledgement of seg (x+6) is received?
6) TCP receiver sometimes waits for 500ms before sending an acknowledgement of a newly arrived segment. For how long will receiver wait before sending ack for seg (x+4)?
7) Assuming seg (x) to be sent in the first transmission round, how many segments will be sent in the third transmission round?
8) Assume no loss occurs, what will be the last segment that will be sent in the *slow start* phase starting from seg(x)?
9) What is the link utilization during the *slow start* phase if the link capacity is 10 Mbps and the RTT between node A and B is 15ms?
10) Suppose after receiving 50 segments from the source, node B lost synchronization with A. Write the <u>name and value of the field</u> used by destination B to notify source node A.

NOTE: Answer all numbers in decimal number system ONLY (where applies). For segment numbers, use the notation of seg(x) where *x* is the number of a segment.



#	Answers	Points
1	**(A) 4578 (B) 4386**	2
2	**(A) Starting byte number: 4386 Ending byte number: 4449**	2
3	**44**	2
4	**300**	2
5	**Window size: 6 ssthreshold: 6**	4

6	It will not wait and send ack 4706	2
7	4	2
8	Seg(x+12)	2
9	(A) Formula used = $\frac{L/R}{RTT+L/R}$ Input values: $\frac{13*64*8/10Mbps}{15ms+64*8/10Mbps}$	2
	(B) Final answer after calculation = 0.044.	2
10	RST bit = 1	2

SOLUTION

0014

National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Computer Networks	Course Code:	CS3001
Degree Program:	BCS, BSE	Semester:	Spring 2023
Exam Duration:	180 Minutes	Total Marks:	80
Paper Date:	31-May-2023	Weight	45%
Section:	ALL	Page(s):	11 + 1(Rough Page)
Exam Type:	Final		

Name:_______________________ Roll No.___________________ Section:__________

Instruction/Notes:
- Attempt all questions on the provided space in the question paper.
- Space for rough work is provided at the end of the paper.
- Even if you do use rough sheets, they should NOT be attached with final paper.
- If you find any ambiguity in a question, you can make your own assumption and answer the question accordingly by stating your assumption.

Question #	1	2	3	4	5	6	7
Total Marks	10	10	10	10	15	10	15
Obtained Marks							
CLO #	1	1	2	3	3	3	3

Question 1: Answer the following multiple-choice questions by **filling the following table**. Cutting and overwriting is not allowed. Any answer outside the table will be awarded **zero marks**. [1+1+1+1+1+1+1+1+1+1 = 10 Marks] (CLO 1)

Any answers outside the table will NOT be marked.

1	B
2	C
3	A
4	D
5	A
6	A
7	C
8	B
9	C
10	B

1.1. Which of the following statement(s) is true

A. Clients are part of the network edge, while Servers and Routers are part of the network core

B. Clients and Servers are part of the network edge, while Routers are part of the network core

C. Clients, Servers and Routers are all part of the network core

D. Clients, Servers and Routers are all part of the network edge

1.2. Following is/are the similarity/similarities between Routers & Switches

A. Both are Layer 3 devices

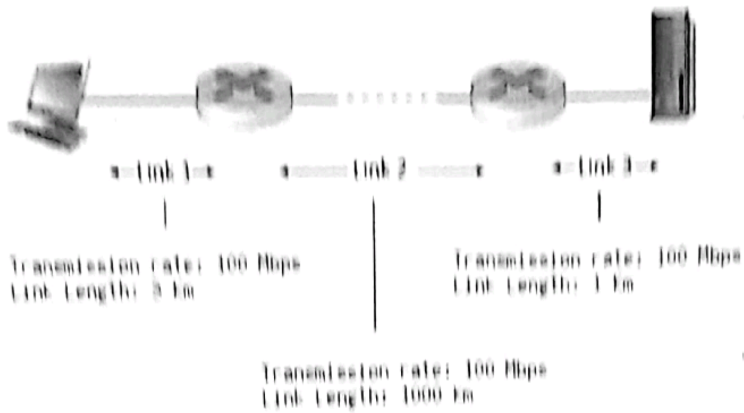
B. Both are transparent to the end systems

C. Both are store & forward and have forwarding tables

D. They have no similarities

8

Question 2: Consider the figure below, with three links, each with the specified transmission rate and link length **[10 Marks] (CLO 1)**



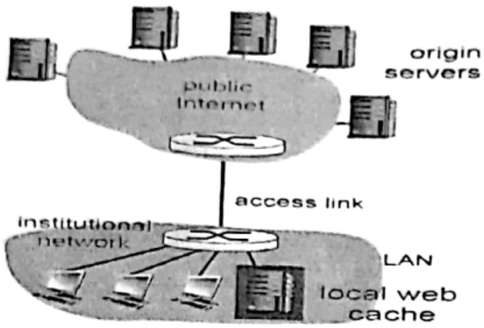
Assume the length of a packet is 8000 bits. The speed of light propagation delay on each link is 3×10^8 m/sec.

Calculate the following Delays:

	Link 1	Link 2	Link 3
Transmission Delay	L/R = 8000 bits / 100 Mbps = 8.00×10^{-5} seconds	= 8000 / 100 = 8×10^{-5} seconds	= 8000 / 100 = 8×10^{-5} seconds
Propagation Delay	$d/s = \frac{(3\ km)*1000\ m}{3*10^8\ s}$ = 1×10^{-5} seconds	= $\frac{1000*1000}{3*10^8}$ = 0.0033 seconds	= $\frac{1*1000}{3\times10^8}$ = 3.33×10^{-6} sec
Total Delay	= $8\times10^{-5} + 1\times10^{-5} = 9\times10^{-5}$	= 8×10^{-5} + 0.0033 = 0.0034	= $8\times10^{-5} + 3.33\times10^{-6}$ = 8.33×10^{-5}

Total Delay of the path = 9×10^{-5} + 0.0034 + 8.33×10^{-5} = 0.0036 seconds

Question 3: Refer to the figure below, a web cache (proxy server) can significantly reduce the overall average delay to the internet (in the case of a cache hit), but there may be an increase in this total average delay (in the case of a cache miss.) Following are the total delays on average (totals including both directions). Let the total LAN delay be 15 ms, the total access link delay be 55 ms, the total internet delay be 300 ms and the total cache penalty delay be 25 ms (where cache penalty delay is the time required to check the cache in both the scenarios, i.e. either a cache hit or a cache miss.) Ignore all other delays. **[1 + 2 + 2 + 5 = 10 Marks] (CLO 2)**



(a) Find the total delay in the scenario where no cache is used? [1]

Total Delay = LAN Delay + Access Link Delay + Internet Delay = 15ms + 55ms + 300ms
= 370ms

(b) Find the total delay when a cache is used with a hit ratio of 90%? [2]

= 90% * (LAN Delay + Cache Penalty) + 10% (LAN Delay + Cache Penalty + Access Link Delay + Internet Delay)

= 0.9 * (15+25) + 0.1 * (15+25+55+300) = 36 + 39.5 = 75.5 ms

(c) Find the total delay when a cache is used with a hit ratio of 10%? [2]

= 10% * (LAN Delay + Cache Penalty) + 90% (LAN Delay + Cache Penalty + Access link Delay + Internet Delay)

= 0.1 * (15+25) + 0.9 * (15+25+55+300) = 4 + 355.5 = 359.5 ms

(d) Find the hit ratio above which the use of cache is justified? [5]

Let y = hit ratio, cache's use is justified when total delay including cache penalty is less than the total delay with using cache. i.e less than 370 ms.

y * (Lan Delay + Cache Penalty) + (1-y) * LAN Delay + Cache Penalty + Access Delay + internet delay) < 370ms.

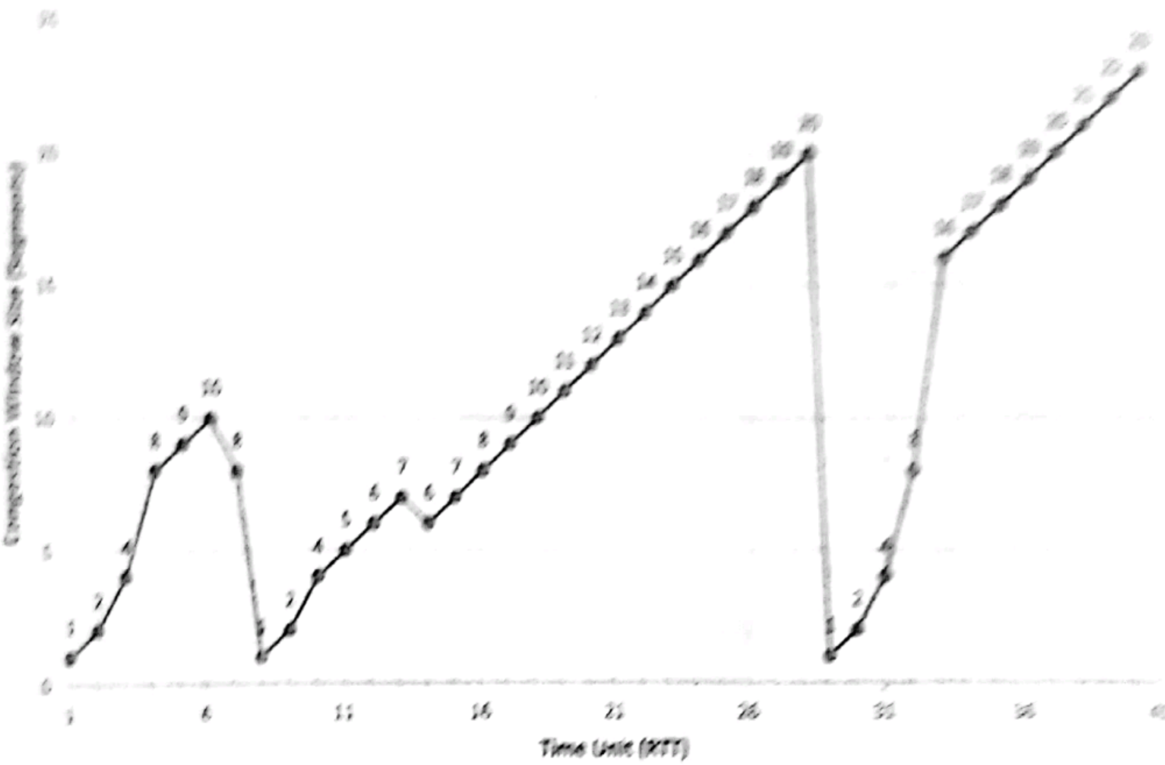
⇒ y * (15+25) + (1-y) * (15+25+55+300) < 370

Solving for y gives

y > 0.0704

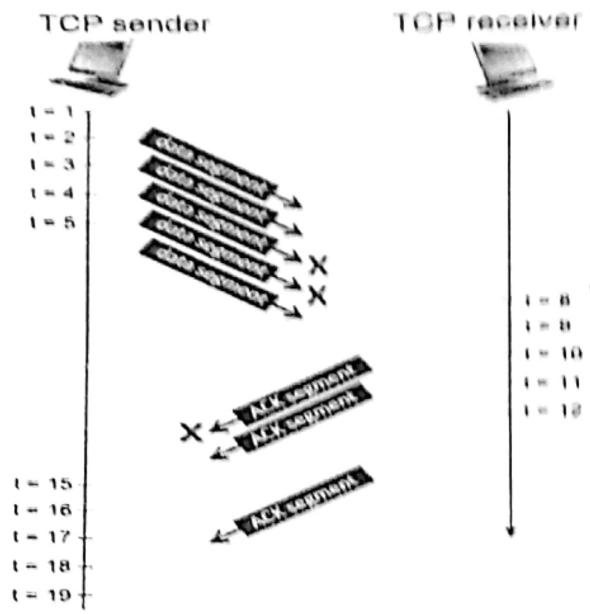
$\boxed{\text{or } 7.04\%}$

Question 4 (a). Consider the figure below, which plots the evolution of TCP's congestion window at the beginning of each time unit (where the unit of time is equal to the RTT). In the abstract model for this problem, TCP sends a "flight" of packets of size cwnd at the beginning of each time unit. The result of sending that flight of packets is that either (i) all packets are ACKed at the end of the time unit, (ii) there is a timeout for the first packet, or (iii) there is a triple duplicate ACK for the first packet. In this problem, you are asked to reconstruct the sequence of events (ACKs, losses) that resulted in the evolution of TCP's cwnd shown below. The initial value of cwnd is 1 and the initial value of ssthresh is 8. (values of x-axis, i.e. RTT, starts from 1) (5 + 5 = 10 Marks) (CLO 3)



Mention the RTT intervals during which TCP is in slow start phase	1-4, 8-10, 29-33
The times at which congestion avoidance phase started.	4, 10, 14, 33
The times at which packets are lost via timeout.	7, 28
The times at which the value of ssthresh changes.	7, 8, 14, 29
The times at which packets are lost via triple ACK.	6, 13

(b): Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at t = 1, 2, 3, 4, and 5, respectively. Suppose the initial value of the sequence number is 187 and every segment sent to the receiver each contains 550 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at t = 8, and an ACK for this segment arrives at t = 15. As shown in the figure, 2 of the 5 segments is lost between the sender and the receiver, but one of the ACKs is lost. Assume there are no timeouts and any out of order segments received are thrown out.



If there is no segment to send at a given time, write None.

Sequence numbers for segments at Time:	
2	187+550 = 737
3	737 +550 =1287
5	1287+550 = 1837
16	1837+55 =2387
18	None

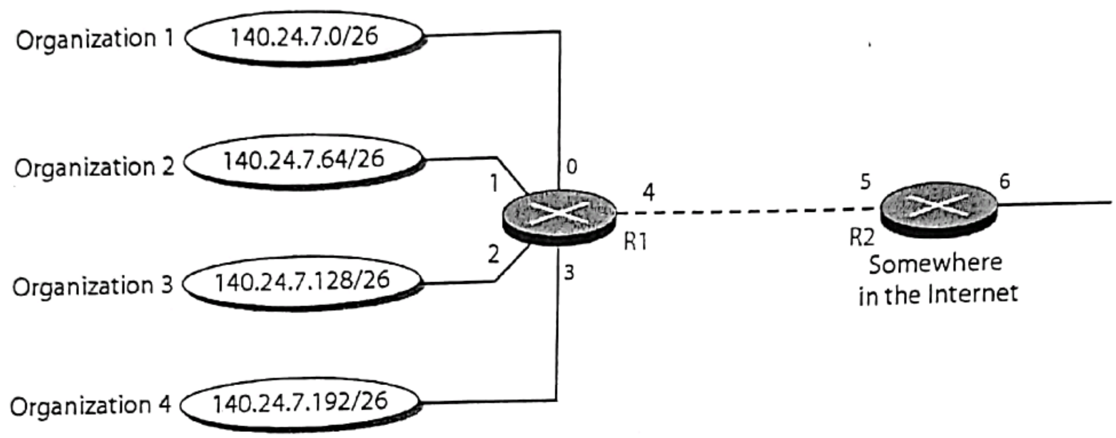
Acknowledgment Number of segments at time:	
8	737
9	1287
10	None
11	None
12	1287

00

Question 5: Answer both parts **(a)** and **(b)**: **[6 + 9 = 15 Marks] (CLO 3)**

(a) Refer to the network scenario in figure 1 below, Routers R1 & R2 with interfaces 0,1,2,3,4 and 5,6 respectively employ IP Address Aggregation (Route Summarization). Please fill in the entries of the tables for both the routers R1 & R2 accordingly. (First entries, i.e. default entries have been pre-filled.) [6]

Figure 1

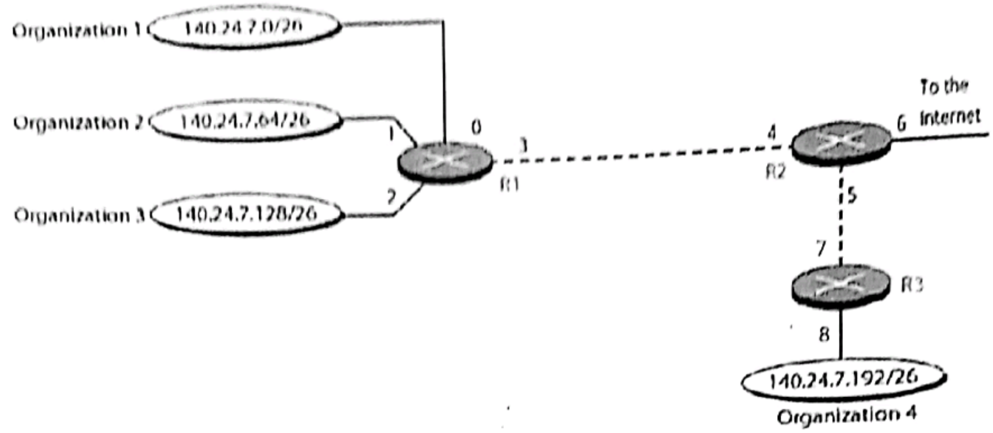


Routing Table for R1	
Network (IP) Address Range with Prefix	Interface
0.0.0.0/0 (Default)	4
140.24.7.0/26	0
140.24.7.64/26	1
140.24.7.128/26	2
140.24.7.192/26	3

Routing Table for R2	
Network (IP) Address Range with Prefix	Interface
0.0.0.0/0 (Default)	6
140.24.7.0/24	5

(b) Now refer to network scenario in figure 2 below, Routers R1, R2 and R3 with their respective interfaces 0,1,2,3........ etc. employ IP Address Aggregation as well as Longest Prefix matching. Please fill in the entries of the tables for the routers R1, R2 & R3 accordingly. (First entries, i.e. default entries have been pre-filled.) [9]

Figure 2



Routing Table for R1

Network (IP) Address Range with Prefix	Interface
0.0.0.0/0 (Default)	3
140.24.7.0/26	0
140.24.7.64/26	1
140.24.7.128/26	2

Routing Table for R2

Network (IP) Address Range with Prefix	Interface
0.0.0.0/0 (Default)	6
140.24.7.0/26	5
140.24.7.0/24	4

Routing Table for R3

Network (IP) Address Range with Prefix	Network (IP) Address Range with Prefix
0.0.0.0/0 (Default)	7
140.24.7.192/26	8
140.24.7.0/24	7

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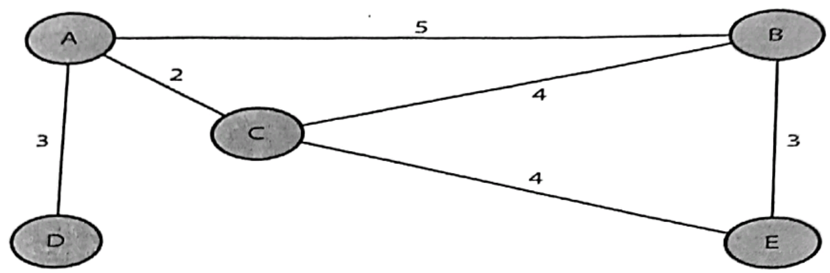
Question 6: Answer both parts **(a)** and **(b)**: [6 + 4 = 10 Marks](CLO 3)

(a) Refer to the autonomous system in the figure below, OSPF is running in all the routers of this AS. Please compute the shortest path from Router A to all the other Routers B, C, D & E. [6]



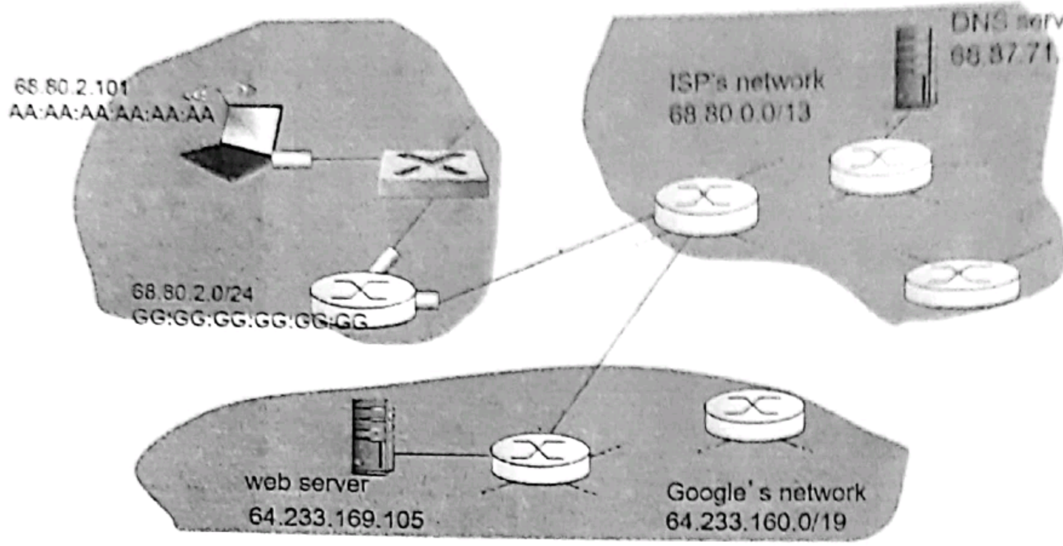
Step	N'	D(B) p(B)	D(C) p(C)	D(D) p(D)	D(E) p(E)
0	A	5, A	2, A	3, A	∞
1	AC	5, A		3, A	6, C
2	ACD	5, A			6, C
3	ACDB				6, C
4	ACDBE				

(b) Consider an AS with routers A, B, C, D & E with the topology **A—B—C—D—E** where hop-count is the metric. A goes down. B does not receive the vector update from A so it concludes that its route of cost 1 to A is no longer available. C doesn't know yet that A is down and tells B that A is 2 hops away from it. Thus the wrong info propagates until it reaches infinity. Briefly give the solution to this problem? **[4]**

Answer:

Poison Reverse. i.e. C advertise to B, the route to A with a cost of infinity, so that B does not route to A via C.

Question 7: Suppose you walk into an organization, connect your laptop to Ethernet, and want to download a Web page. What are all the protocol steps that take place, starting from powering on your PC to getting the Web page? Assume there is nothing in our DNS or browser caches when you power on your PC. [15 Marks] (CLO 3)



First Type of Packet Sent from your host is	DHCP request message	
	Source	Destination
Port Number	68	67
IP Address	0.0.0.0	255.255.255.255
MAC Address	AA.AA.AA.A.A.AA.AA	FF.FF.FF.FF.FF.FF
Application Layer Protocol	DHCP	
Transport Layer Protocol	UDP	

First Packet Received on your host is: DHCP ACK message

It contains: (Fill in the missing items in either of the columns in the table below.)

1) Alloted IP address for your host	68.80.2.101
2) IP address of DNS server	68.87.71.226
3) IP address of gateway router	68.80.0.2
4) Subnet Mask	/24

How many messages/packets are exchanged in getting IP address for your host? 4.

Next ARP query message will be generated to get Mac Address of gateway router

ARP Query	Source	Destination
IP Address	68.80.2.101	68.80.2.0
MAC Address	AA:AA:AA:AA:AA:AA	FF:FF:FF:FF:FF:FF

ARP response message will contain:

	Source	Destination
IP Address	68.80.2.0	68.80.2.101
MAC Address	GG:GG:GG:GG:GG:GG	AA:AA:AA:AA:AA:AA

After getting IP address, and ARP response next packet sent from your host is		DNS Query Message
	Source	Destination
Port Number		53
IP Address	68.80.2.101	68.87.71.226
MAC Address	AA:AA:AA:AA:AA:AA	GG:GG:GG:GG:GG:GG
Application Layer Protocol	DNS Protocol	
Transport Layer Protocol	UDP	

As a response of the above query your host will get: DNS reply message (host name to IP mapping of web server)

Now is the time to contact webserver for your required webpage which uses HTTP.
How many packets will be exchanged between your host and server before sending actual data and why?

2, HTTP uses TCP which uses 3-way handshake for establishing connection, before sending actual data as payload in 3rd packet.

National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Computer Networks	Course Code:	CS3001
Degree Program:	BS (CS), BS (SE), BS (DS), BS (Robotics)	Semester:	Fall 2023
Exam Duration:	180 Minutes	Total Marks:	85
Paper Date:	02-January-2024	Weight	45%
Section:	ALL	Page(s):	11 + 1 (Rough Page)
Exam Type:	Final		

Name: [redacted] Roll No. [redacted] Section: [redacted]

Instruction/Notes:
- Attempt all questions on the provided space in the question paper.
- Space for rough work is provided at the end of the paper.
- Even if you do use rough sheets, they should NOT be attached with final paper.
- If you find any ambiguity in a question, you can make your own assumption and answer the question accordingly by stating your assumption.

Question #	1	2	3	4	5	6	7	8	
Total Marks	10	8	6	6	9	10	20	16	85
Obtained Marks	7	8	0	5	9	7	17	14.5	67.5
CLO #	1	2	2	2	3	3	3	~~4~~ 3	

Question 1: Answer the following multiple-choice questions by **filling the following table**. Cutting and overwriting is not allowed. Any answer outside the table will be awarded **zero marks**. [1+1+1+1+1+1+1+1+1+1 = 10] (CLO 1)

Any answers outside the table will NOT be marked.

1.1	B ✓
1.2	B ✓
1.3	C ✗
1.4	A ✗
1.5	C ✓
1.6	D ✓
1.7	D ✗
1.8	B ✓
1.9	D ✓
1.10	B ✓

7

1.1. Time needed to perform an integrity check, lookup packet information in a local table and move the packet from an input link to an output link in a router.

A. Queueing delay
B. Processing delay
C. Transmission delay
D. Propagation delay

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1.2. True or False? The fraction of lost packets increases as the traffic intensity decreases.
A. True
B. False

1.3. Urg data pointer field in TCP segment consists of __________ bits:
A. 8
B. 16
C. 32
D. 64

1.4. The sending rate in TCP is adjusted according to the:
A. Congestion Window (cwnd)
B. Receive Window (rwnd)
C. Minimum of (cwnd , rwnd)
D. Urgent Data Pointer

1.5. What is the maximum length of IPv4 Header?
A. 20 Bytes
B. 40 Bytes
C. 60 Bytes
D. 80 Bytes

1.6. How does a host get it's own IP address & get the IP address of another host across the internet?
A. via DHCP & ARP respectively
B. via DNS & ARP respectively
C. via DNS & DHCP respectively
D. via DHCP & DNS respectively

1.7. The notification message in BGP can indicate:
A. Error in previous message
B. Close connection
C. Both A and B
D. None of the above

1.8. Ethernet's MAC protocol is based on:
A. Token Passing
B. CSMA / CD
C. CSMA / CA
D. Both B and C

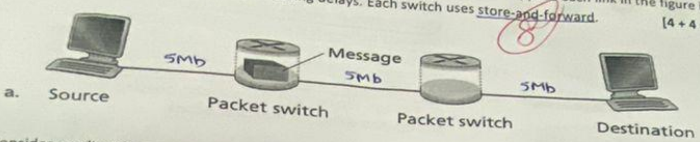
1.9. To maintain cookies, what components are needed other than the cookie file on the user's host:
A. Cookie header line of the HTTP response message
B. Backend database at the website
C. Cookie header line in the next HTTP request message
D. All of the above

1.10. HTTP 2 mitigates __________ by interleaving frame transmission:
A. Performance
B. Head Of Line (HOL) blocking
C. Total Transfer Time
D. Transfer Time for Large Objects

Question 2: In packet-switched networks, like the Internet, the source host segments long, application-layer messages (e.g. an image or a music file) into smaller packets and sends the packets into the network. The receiver reassembles the packets back into the original message. This process is known as message segmentation. The below figure illustrates the end-to-end transport of a message with and without message segmentation. Consider a message that is 10^6 bits long that is to be sent from source to destination. Suppose each link in the figure is 5 Mbps. Ignore propagation, queuing, and processing delays. Each switch uses store-and-forward. [4 + 4 = 8] (CLO 2)

8

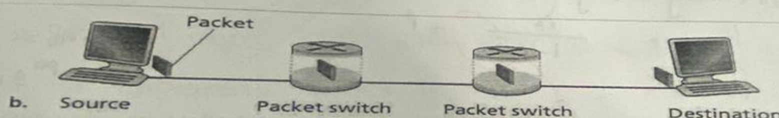
a.



Consider sending the message from source to destination without message segmentation [4]

How long does it take to move the message from the source host to the first packet switch?
5mb = $\frac{10^6}{5\times10^6}$ = 0.2sec 4
What is the total time to move the message from source host to destination host?
We have to go through 3 hops so 3(0.2) => 0.6sec

b.



Now suppose that the message is segmented into 100 packets, with each packet being 10,000 bits long. [4]

How long does it take to move the first packet from source host to the first switch?
Now => $\frac{10{,}000}{5\times10^6}$ = 0.002 sec (ignoring all other delays) 4
At what time will the second packet be fully received at the first switch?
Since its store & forward, second packet will be sent after the first packet is reached the first switch so The Time will be = $\boxed{0.004sec}$

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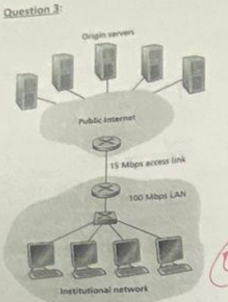
Question 3:



Consider this figure on the left.

There is an institutional network connected to the Internet. Suppose that the average object size is 850,000 bits and that the average request rate from the institutions' browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is 3 seconds on average. *You can ignore processing, queuing, and propagation delays for your calculations.*

Note that the total average response time (delay) as the sum of the average LAN delay, average access delay (that is, the delay from the Internet router to the institutional router) and the average Internet delay. For the average access delay, use $\Delta/(1-\Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the request rate per second.

0

[3]

a) Find the average response time ignoring LAN and access link propagation delays.

Avg obj size = 850,000 bits = 0.8 Mb
β = 16 req per second
Δ = 3 sec

= LAN delay + Avg Access delay + Internet Delay

$= \int\int + \dfrac{\Delta}{1-\Delta\beta} + \beta$ ≈ 0.18 sec per 1 req

= 15.93

b) Find the total response time for all 16 objects (requests per second) considering that a web cache is installed in the institutional LAN (**with a miss rate of 0.4**). Ignore LAN and access link propagation delays and assume that there is no cache penalty.

[3]

miss rate = 0.4

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Question 4: [2 + 2 + 2 = 6] (CLO 2)

5

i. Within the distributed database of resource records, DNS stores a four-tuple, each record has a *name* field, *value* field and *type* field. What is the other field, and describe its use?

Answer:

TTL → Time to Live, it is the 4th field which is stored at DNS. It shows us the time after which that particular (info about domain) will be dropped

2

ii. For records with *type*=NS, we know the *name* field stores the domain, whereas the *value* field stores the authoritative name server for that domain. What is stored in the *name* and *value* fields for records with *type*=CNAME?

Answer:

In CNAME type,
name = Original Name (Domain)
value = Alias, Canonical Name

1

iii. If we want to implement DNS in a way that decreases the load at the upper levels of the hierarchy, we would prefer to use recursive DNS. Do you agree with this statement, explain your answer?

Answer:

No, I dont agree with this statement. If we want to implement DNS in a way that decreases the load on upper level of hierarchy, We should use Iterative Approach since it involves local domain iteratively asking at every level whereas recursion would put pressure on upper hierarchy.

2

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Question 5: Suppose that the three measured SampleRTT ... *EstimatedRTT* after each of these SampleRTT values is obtained, ... value of *EstimatedRTT* was 102 ms just before the first of these three samples ... assuming a value of β = 0.25 and assuming the value of *DevRTT* was 4 ms ... *DevRTT* after each sample is obtained, ... Last, compute the TCP *TimeoutInterval* after each of these samples is obtained. before the first of these three samples was obtained. [3 + 3 + 3 = 9]

Answer:

9

Sample RTT = 105ms, 110, 115ms

$\alpha = 0.125$, Estimated RTT = 102ms

$\beta = 0.25$, Dev RTT = 4ms

Estimated RTT = $(1-\alpha)$ Estimated RTT + α Sample RTT

for 105

ERTT = $(1-0.125)102 + 0.125(105\text{ms})$

= 89.25 + 13.125

$\boxed{ERTT_1 = 102.375}$

for 110

$ERTT_2 = (1-0.125)(102.375) + 0.125(110)$

89.58 + 13.75

$\boxed{ERTT_2 = 103.328}$

for 115

$ERTT_3 = (1-0.125)(103.328) + 0.125(115)$

= 90.412 + 14.375

$\boxed{ERTT_2 = 104.787}$

Dev RTT = $(1-\beta)$ Dev RTT + β |estimated RTT − Sample RTT|

for 105

= $(1-0.25)\,4 + 0.25|102.375 - 105|$

= 3 + 0.65625

$\boxed{DevRTT_1 = 3.65625}$

for 110

$DevRTT_2 = (1-0.25)(3.65625) + 0.25|110 - 103.328|$

= 2.74 + 1.668

$\boxed{DevRTT_2 = 4.41}$

for 115

$DevRTT_3 = (1-0.25)(4.41) + 0.25|115 - 104.787|$

= 3.3075 + 2.55325

$\boxed{DevRTT_3 = 5.86}$

Timeout = Est RTT + 4×D

for

105 = 102.375 + 4(3.65

$\boxed{1st = 117}$

for

110 = 103.328 + 4(4.4

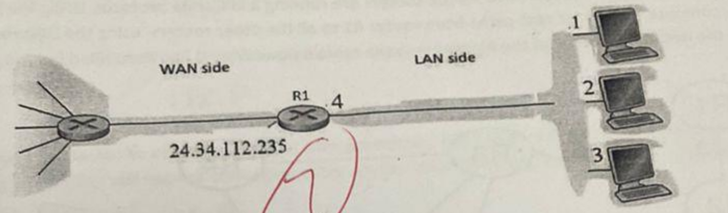
$\boxed{2^{nd} = 120.968 \approx 121}$

for

115 = 104.787 × 5.86(4

$\boxed{3^{rd} = 128.227}$

Question 6: Consider the network setup in the following figure. Suppose that the ISP assigns the router R1 the address 24.34.112.235 on the WAN side. Also suppose that the subnet address of the home network is 192.168.1.0/24. [4 + 6 = 10] (CLO 3)



2

a) Assign addresses to all interfaces numbered 1 to 4 in the home network.

IP address of host at interface-1: 192.168.1.1/24

IP address of host at interface-2: 192.168.1.2/24

IP address of host at interface-3: 192.168.1.3/24

IP address of router R1 at interface-4: 192.168.1.0/24

3

b) Suppose each host has two ongoing TCP connections, all to port 80 at a host 128.119.40.86 residing somewhere in the WAN side (Internet). For that purpose, the host at interface-1 uses source ports 3345 and 3346; the host at interface-2 uses source ports 3355 and 3356; and the host at interface-3 uses source ports 3365 and 3366. Also suppose that the router R1 uses source port numbers greater than 4120 to establish TCP connections. Provide the six corresponding entries in the NAT translation table.

This will be how they would be shown at other world

These will be the local unique address of each host

4

NAT Translation Table

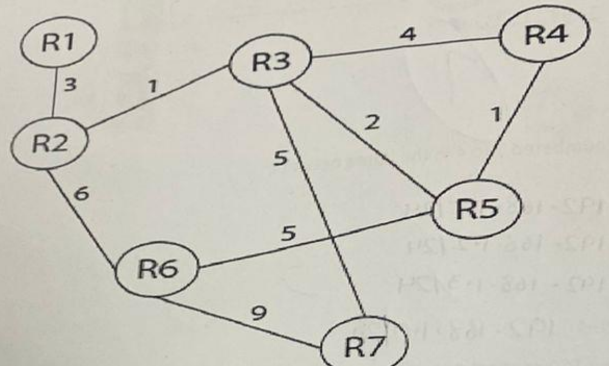
WAN Side	LAN Side	
192.168.1.0, 4121	192.168.1.1, 3345	will be sending to 128.119.40.86
192.168.1.0, 4122	192.168.1.1, 3346	→ 128.119.40.8
192.168.1.0, 4123	192.168.1.2, 3355	→ 128.119.40
192.168.1.0, 4124	192.168.1.2, 3356	→ 128.119.4
192.168.1.0, 4125	192.168.1.3, 3365	→ 128.119.40.
192.168.1.0, 4126	192.168.1.3, 3366	→ 128.119.40

[13 + 7 = 20

(b) Answer al
i) Whic

Question 7: Please answer both parts **(a) & (b)**

(a) Refer to the figure below, it contains a network topology containing routers R1 till R7 with the link cost mentioned against each link. All the routers are running a link state protocol. Using the table below, please compute the lowest cost paths from router R1 to all the other routers, using the Dijkstra's algorithm. List all the iterations (steps) of the algorithm in the table below. (Step 0 has been filled for you.) [13] (CLO



17

13

Step	N'	R2 D(R2), p(R2)	R3 D(R3), p(R3)	R4 D(R4), p(R4)	R5 D(R5), p(R5)	R6 D(R6), p(R6)	R7 D(R7), p(
0	R1	3 , R1	∞	∞	∞	∞	∞
1	R1,R2	3,R1	4,R2	∞	∞	9,R2	∞
2	R1,R2,R3	3,R1	4,R2	8,R3	6,R3	9,R2	9,R3
3	R1,R2,R3,R5	3,R1	4,R2	7,R5	6,R3	9,R2	9,R
4	R1,R2,R3,R5,R4	3,R1	4,R2	7,R5	6,R3	9,R2	9,R
5	R1,R2,R3,R5,R4,R6	3,R1	4,R2	7,R5	6,R3	9,R2	9,R3
6	all	3,R1	4,R2	7,R5	6,R3	9,R2	9,R3

R1,R2, R3,R5, R6,R7

4

(b) Answer all the parts below: (Please note IGP is the same as Intra-AS Routing Protocol) [7] (CLO 3)

i) Which of the three (IGP, iBGP, eBGP) carry intra-domain (or intra-AS) routing information?

Answer: 1 IGP

ii) Which of the three (IGP, iBGP, eBGP) carry inter-domain (or inter-AS) routing information?

Answer: 1 IBGP

iii) A host sends a packet to another within its own domain (or AS) but on a different LAN/subnet. Correctly sending this packet will require consulting table entries created by which of the three (IGP, iBGP, eBGP)?

Answer: 1 IGP

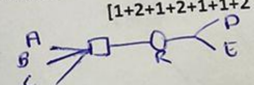
iv) A host sends a packet to another host in a different domain (or AS). Along the way, correctly sending this packet will require consulting table entries created by which of the three (IGP, iBGP, eBGP)?

Answer: eBGP 1

14.5

Question 8: Answer the following questions: **[10 + 6 = 16] (CLO 3)**

I) Consider three stations A, B, and C are on the same extended LAN. Two other stations D and E are part of a different LAN connected to the first LAN through a router R. Suppose that all stations know the IP addresses of each other (either on the same LAN or on different LAN) and ARP cache of A, B and C as well as D and E is empty. The following events happen in that order: **[1+2+1+2+1+1+2 = 10 Marks] (CLO 3)**



1. A sends an IP datagram to C.
2. B sends an IP datagram to D.
3. A sends a second IP datagram to C.
4. E sends an IP datagram to B.

9

Answer the following considering the given scenario and sequence of events:

(a) In step (1) above, does station A need to use ARP? (A YES/NO answer is sufficient.)

Answer: NO 6

(b) In step (1), if A launches an ARP request, what should be the target IP address in that request?

Answer: Local address of ~~[illegible]~~ C

(c) In step (2) above, does station B need to use ARP? (A YES/NO answer is sufficient.)

Answer: YES

(d) In step (2), if B launches an ARP request, what should be the target IP address in that request?

Answer: D's local IP address & E's too

(e) In step (3) above, does station A need to use ARP? (A YES/NO answer with justification.)

Answer: NO, since its in its own LAN & the IP is already known

(f) In step (4) above, does station E need to use ARP? (A YES/NO answer is sufficient.)

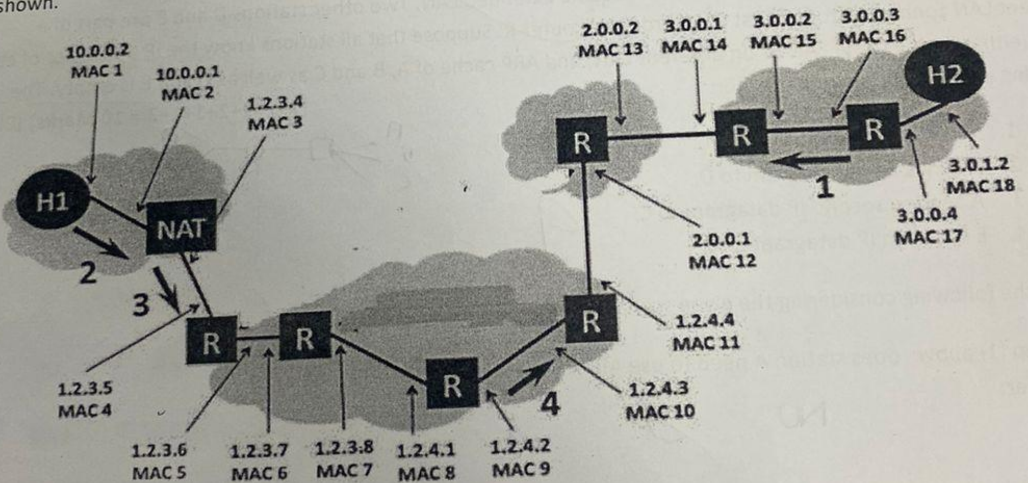
Answer: YES

(g) In step (4), if E launches an ARP request, what should be the target IP address in that request?

Answer: B's Local IP address

II) The figure below shows a network topology. The LAN on the left uses a NAT to connect to the Internet and includes a client host H1. The LAN on the right includes a webserver H2. Packets between the two endpoints are routed along the path shown by heavy dark lines. The various network interfaces have IP and MAC addresses as shown.

[2+2+2=6] (CLO 3)



H1 has established an HTTP session with web server H2 and data packets are flowing between the two machine an example, headers for packet 1 have been filled (traveling from the server H2 to the client H1). Note that you should order your headers from "outermost" in, as shown: Ethernet should be listed before IP, because the Eth packet exists first on the wire.

You have to fill in the header type and the source and destination address for the network and datalink layer headers for packets 2, 3, and 4 (these packets are all traveling from the client H1 to the server H2, as marked on the figure with heavy black arrows and numbers).

5.5

A. Header for packet 1

Header Type	Source	Destination
Ethernet	MAC 16	MAC 15
IP	3.0.1.2	1.2.3.4

B. Header for packet 2

Header Type	Source	Destination
Ethernet	MAC 1	MAC 2
IP	10.0.0.2	3.0.1.2

C. Header for packet 3

Header Type	Source	Destination
Ethernet	MAC 3	MAC 4
IP	1.2.3.4	3.0.1.2

D. Header for packet 4

Header Type	Source	Destination
Ethernet	MAC 9	MAC 10
IP	1.2.4.2	3.0.1.2

Computer Networks (CS3001)

Final Exam

Date: May 27th 2024

Course Instructor(s)

Mr. Nauman Moazzam Hayat

Ms. Umm-e-Ammarah

Total Time: 3 Hours

Total Marks: 80

Total Questions: 8

Semester: SP-2024

Campus: Lahore

Dept: Computer Science / Software Engineering

Student Name ______ Roll No ______ Section SE-6C Student Signature ______

Instruction/Notes:
- Attempt all questions on the provided separate answer sheet.
- Clearly write corresponding question number and part number at the top centre of the answer sheet with a thick pen / marker before starting a new question / answer.
- Fill in the corresponding tables for the relevant questions in the additional sheet provided which is named "<u>Figures & Tables</u>" and attach it with the answer sheet.
- In case, you use rough sheets, they should **NOT** be attached to the final answer sheet.

CLO 1:

Q1: Answer the following multiple-choice questions by filling **Table 1** in the additional sheet named "<u>Figures & Tables</u>": **[1 * 10 = 10 Marks]**

<u>Note: Any answers outside the table will NOT be marked. Moreover, cutting and overwriting is not allowed.</u>

1.1. ________ transfers messages from senders' mail servers to the recipients' mail servers.

A. Simple Network Management Protocol
B. Hyper Text Transfer Protocol
C. Simple Mail Transfer Protocol
D. Transmission Control Protocol

1.2. Elastic Applications are Apps that

A. Are highly customizable for each user
B. Require a minimum throughput to work
C. Make use of whatever throughput they get
D. None of the above

1.3. The sending rate in TCP is adjusted according to the:

A. Congestion Window (cwnd)
B. Receive Window (rwnd)
C. Minimum of (cwnd , rwnd)
D. Urgent Data Pointer

1.4. The Default Subnet Mask for a Class C IP address is

A. 255.255.255.255
B. 255.255.255.0
C. 255.255.0.0
D. 255.0.0.0

1.5. If an AS learns through ________ protocol that subnet x is reachable from more than one other ASes, choosing the gateway that has the smallest __________ is called ____________.

A. inter-AS; least cost; hot potato routing
B. inter-AS; least cost; cold potato routing
C. intra-AS; least cost; hot potato routing
D. intra-AS; average cost; hot potato routing

1.6. Although a node senses the carrier before transmitting, collisions can still occur in CSMA because

A. of propagation delay
B. of transmission delay
C. of processing delay
D. collisions do not occur in CSMA

1.7. Time needed to perform an integrity check, lookup packet information in a local table and move the packet from an input link to an output link in a router.

A. Queueing delay
B. Processing delay
C. Transmission delay
D. Propagation delay

1.8. A node can get it's own IP address : IP address of another node on the internet : MAC Address of a node on the same LAN respectively via

A. DHCP : ARP : DNS
B. DNS : DHCP : ARP
C. ARP : DNS : DHCP
D. DHCP : DNS : ARP

1.9. A switch having 7 interface, will have:

A. 7 MAC addresses
B. 5 MAC addresses
C. 1 MAC address
D. No MAC address

1.10. One similarity : One Difference between a switch & a router are: (Please choose only one option from below for which both the statements are correct)

A. Both switch and router are plug & play : Switch is a Layer 2, while Router is a Layer 3 device
B. Both have forwarding tables : Switch participates in local delivery while routers in global delivery
C. Both use store and forward : Switch uses broadcasting while router uses flooding
D. None of the above option is entirely true

CLO 2:

Q2: Refer to **Figure 2** in the additional sheet named "**Figures & Tables**", suppose a caravan has 20 cars, and the tollbooth services a car at a rate of one car per 2 seconds. Once serviced, a car proceeds to the next toll booth, which is 400 kilometers away at a rate of 10 kilometers per second. The entire caravan must be queued at a tollbooth before the first car in the caravan can be serviced (i.e. pay its toll and begin driving towards the next tollbooth.)

Answer all the parts a, b, c , d and e in the answer sheet. **[2 * 5 = 10 Marks]**

a. How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?

b. Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?

c. Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?

d. Are there ever two cars in service at the same time, one at the first toll booth and one at the second toll booth? And Why?

e. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first toll booth but not yet arrived at the second toll booth? And Why?

CLO 2:

Q3 (a): A user is trying to visit a website www.example.com, but her browser doesn't know the IP address. Using iterative DNS query, answer all parts i till v in the answer sheet. [1 * 5 = 5 Marks]

i) Where does the Local DNS server check first (Where does it send the DNS query first)?
ii) Where does the Local DNS Server check next (Where does it send the DNS query next)?
iii) Where does the Local DNS Server check next (Where does it send the DNS query next)?
iv) What type of DNS Record (RR) is returned in response to this DNS Query?
v) Which type of DNS query is considered best practice (Iterative or Recursive)?

(b) Below is a server to client HTTP Response Message. Answer all parts i till v in the answer sheet: [1 * 5 = 5 Marks]

HTTP/1.0 404 Not Found
Date: Tue, 07 May 2024 11:33:45 +0000
Server: Apache/2.2.3 (CentOS)
Content-Length: 258
Connection: Close
Content-type: text/html

i) Was the server able to send the file (object) successfully?
ii) Is the HTTP connection persistent or non persistent?
iii) What is the type of the file (object) being sent?
iv) What is the size of the file (object) being sent in bytes?
v) What is the name & version of the server?

CLO 2:

Q4. Refer to Figure 4 in the additional sheet named "Figures & Tables, the left and right TCP clients communicate with a TCP server using TCP sockets. The three sockets shown in server were created as a result of the server accepting connection requests on this welcoming socket from the two clients. Fill in Table 4 in the additional sheet named "Figures & Tables." [8 Marks]

CLO 3:

Q5: Refer to Figure 5 in the additional sheet named "Figures & Tables, IPv6 subnets connected by a mix of IPv6-only routers, IPv4-only routers & IPv6/IPv4 routers. A host of subnet D wants to send an IPv6 datagram to a host on subnet B. The forwarding between these two hosts goes along the path: D --> E --> d --> c --> B. Answer all the parts i till x in the answer sheet: [1 * 10 = 10 Marks]

i) Is the datagram being forwarded from D to E an IPv4 or IPv6 datagram?
ii) Is this D to E datagram encapsulating another datagram? Yes or No?
iii) Is the datagram being forwarded from E to d an IPv4 or IPv6 datagram?

iv) Is this E to d datagram encapsulating another datagram? Yes or No?
v) Is the datagram being forwarded from d to c an IPv4 or IPv6 datagram?
vi) Is this d to c datagram encapsulating another datagram? Yes or No.
vii) Is the datagram being forwarded from c to B an IPv4 or IPv6 datagram?
viii) Is this c to B datagram encapsulating another datagram? Yes or No.
ix) Which router is the 'tunnel entrance'?
x) Which router is the 'tunnel exit'?

CLO 3:

Q6 (a): Refer to **Figure 6** in the additional sheet named "**Figures & Tables**, run Dijkstra's Algorithm and fill in **Table 6** in the additional sheet named "**Figures & Tables**." **[6 Marks]**
(b) What are the 4 route selection rules (in order) in BGP, if more than one route is available for a specific destination. Write your answer in the answer sheet: **[1 * 4 = 4 Marks]**

CLO 3:

Q7: Refer to **Figure 7** in the additional sheet named "**Figures & Tables**, consider the LAN consisting of computers connected by two self-learning Ethernet switches. At t = 0, the switch table entries for both switches are empty. At t = 1, 2, 3, 4, 5, 6, 7, 8, and 9, a source sends a frame to a destination as shown below, and the destination replies immediately at the same time t(well before the next time step, for example if source sends at t = 9, the destination also replies at t = 9, i.e. it replies before t = 10.) Answer all the parts i till v in the answer sheet: **[2 * 5 = 10 Marks]**
i) At t = 9, which node sent a frame to which node? (If there is only enough information for 1 node, write that node, if there's no information, write 'n/a')
ii) At t = 6, which node sent a frame to which node? (If there is only enough information for 1 node, write that node, if there's no information, write 'n/a')
iii) At t = 3, which node sent a frame to which node? (If there is only enough information for 1 node, write that node, if there's no information, write 'n/a')
iv) At t = 7, which node sent a frame to which node? (If there is only enough information for 1 node, write that node, if there's no information, write 'n/a')
v) At t = 11, which node sent a frame to which node? (If there is only enough information for 1 node, write that node, if there's no information, write 'n/a')

CLO 3: **[1 * 2 = 2 Marks]**
Q8 (a): Answer both parts (i) & (ii) in the answer sheet:
i) Which of the three (OSPF, iBGP, eBGP) carry intradomain (or intra-AS) routing information?
ii) Which of the three (OSPF, iBGP, eBGP) carry interdomain (or inter-AS) routing information?

(b) Refer to **Figure 8** in the additional sheet named "**Figures & Tables**, consider an IP datagram being sent from node D to node B. Answer all the parts i till v in the answer sheet: **[2 * 5 = 10 Marks]**
i) What is the source mac address at point 5?
ii) What is the destination mac address at point 5?
iii) What is the source IP address at point 5?
iv) What is the destination IP address at point 5?
v) What is the source mac address at point 3?

Computer Networks (CS3001)

Date: December26th,2024

Course Instructor(s)

Dr. Arshad Ali, Dr. Abdul Qadeer
Dr Ahmad Raza, Aftab Alam
Ms. Umm-e-Ammarah, Ms. Saba Tariq
Mr. Nauman Moazzam Hayat

Final Exam

Total Time: 180 Minutes
Total Marks: 80
Total Questions: 08

Semester: Fall-2024
Campus: Lahore
Dept: Computer Science

[redacted] [redacted] [redacted]

Student Name Roll No Section Student Signature

Instruction/Notes:

- Attempt all questions on the provided separate answer sheet.
- You are required to attempt all questions and parts thereof in a sequence. This implies that first attempt all parts of Question 1, then the same for Question 2 and so on.
- If you find any ambiguity in a question, you can make your own assumption and answer the question accordingly by stating the same.
- The question paper with your particulars is required to be attached with the answer sheet. In case you have used rough sheets, they should NOT be attached to the answer sheet.

CLO 1 (Question 1): Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Question # 1: Select and write the correct option for the following multiple-choice questions on the provided answer sheet. Any ***cutting and overwriting will be marked as zero.*** *Moreover, if you encircle or write your option(s) on question paper, the same will not be marked.* **[1 * 8 = 8 Marks]**

1.1. What does it mean for a network to achieve a "high utilization"?
A. The network maintains low latency. B. The network fully uses available bandwidth.
C. Packets are evenly distributed across all routes. D. The network ensures no packet loss occurs.

1.2. What is a distinguishing feature of persistent HTTP?
A. Separate connections are used for transfer of each object.
B. All objects are sent over a single TCP connection.
C. It uses UDP for faster performance.
D. It uses cookies for tracking state.

1.3. Which of the following is NOT true about SMTP?
A. It uses persistent TCP connections B. It requires email headers in 7-bit ASCII format
C. It is used to retrieve emails from the mail server D. It can deliver messages in HTML format

1.4. Which feature of TCP ensures reliable data delivery?
A. Three-way handshake B. Congestion control C. Timeout D. Retransmission of lost segments

1.5. What does a link-state routing algorithm require from all routers?
A. Periodic exchange of full routing tables. B. Knowledge of the entire network topology.
C. Prior establishment of a virtual circuit. D. Packet marking for traffic prioritization.

1.6. Which statement about Ethernet's CSMA/CD protocol is TRUE?

A. Collisions are avoided by reserving the medium in advance.
B. Collisions are detected, and affected frames are retransmitted.
C. CSMA/CD is designed for wireless networks.
D. It ensures fair bandwidth distribution using tokens.

1.7. Which protocol is used to exchange routing information between Autonomous Systems?

A. OSPF B. RIP C. BGP D. ICMP

1.8. Which of the following is TRUE about UDP?

A. It provides flow control to prevent sender overflow
B. It requires an initial handshake to establish a connection
C. It adds minimal overhead to transmitted data
D. It ensures in-order delivery of packets.

Size = 5000 bit (x10)

R = 10 Mbps

S = 2×10^8 m/s

CLO 2 (Questions # 2,3, 4 & 5): Demonstrate **the basics of network concepts using state-of-the-art network tools/techniques.**

Question # 2: Suppose that 10 identical packets each comprising 5000-bits are sent back-to-back from source node to destination node with three routers between them making the path as S->R1-> R2-> R3->D. Thus, there are 4 links in total. Now, suppose that each link has a transmission rate of 10 Mbps, propagation speed for each transmission link is $2*10^8$ m/s while length of each links is 500 km. Consider that each router incurs a per packet processing delay of 2 ms.

Moreover, assume that all packets are well received and buffered by router R1, and the current time is 0 (called T0). Router R1 starts processing of the first packet at T0 and sends packet one after another. Router R2 incurs the given processing delay on each packet and forwards the same toward router R3. For this scenario, answer the following questions assuming that there is neither any packet loss nor any queuing delays: **[5+5+2 = 12 Marks]**

(2a): At what time will each packet be leaving router R1 (T1), at what time will each packet be received by router R2 (T2), at what time will each packet be leaving router R2 (T3) and at what time will each packet be received by router R3 (T4). For the given scenario, write the first 5 entries (row 1 to 5) as per the following sample table format for the first 5 packets (packets 1 to 5). Remember that the initial time (T0) is 0.

Packet #	T1	T2	T3	T4
1				

(2b): Now suppose that the transmission rate of sender dropped to 50% of its original transmission rate after forwarding 5 packets. Considering this condition, write the next 5 entries (row 6 to 10) for the remaining 5 packets (packets 6 to 10) considering the new transmission rate:

(2c): Compare the total delays for a single packet in both cases. What is the percentage increase in delay when the transmission rate is reduced to 50%?

Note: A packet is considered as leaving a node after necessary processing and transmission delays. -> do dobara

Question # 3: Answer the following questions. **[3+1+1+3 = 8 Marks]**

(3a): A client needs IPv4 address for the name *cloud.google.com*. The client uses an iterative DNS resolver (local DNS) for name-to-IP resolution. Assume that DNS resolver has no previously stored DNS data. Write the steps taken by the resolver to get IPv4 for the name cloud.google.com

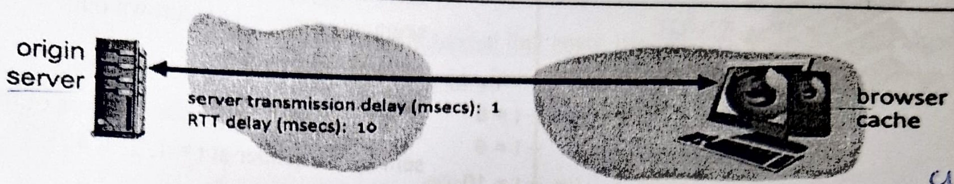
(3b): Two CN students concurrently resolved the name www.google.com. Both students received different IPs. The first student thinks that DNS resolution didn't properly work because same name should have same IP. The second student thinks that Google uses a unique IP for each of its clients. Who is right? First student or the second one?

6 0.012
7 0.014
8 0.016
9 0.018
10 0.02

0.023

(3c): What will be the record type for the name-to-IPv4 mapping in DNS data?

(3d): Consider an HTTP server and client as shown in the figure below. Suppose that the RTT delay between the client and server is 10 msecs; the time a server needs to transmit an object into its outgoing link is 1 msecs; and any other HTTP message not containing an object has a negligible (zero) transmission time. Suppose the client makes 70 requests, one after the other, waiting for a reply to a request before sending the next request. Assume that all of the 70 objects have same size and an object can fully fit in a TCP segment. Also assume that server and client have already established a TCP connection.



Assume the client is using HTTP 1.1 and the IF-MODIFIED-SINCE header line is used in each http request. Assume, 50% of the objects requested have NOT changed since the client downloaded them in an earlier session (before these 70 downloads are performed now). → 35 still downloaded.
How much time elapses (in milliseconds) between the client transmitting the first request, and the completion of the last request?

Question # 4: Answer the following questions: **[2+1+1+3 = 7 Marks]**

(4a) Consider many users which share a 1Gb/s link (Note: $1G = 10^9$). When transmitting, each user requires 40Mb/s (Note: $1M = 10^6$).

(I) How many users (at maximum) can be supported if the users share the link using time division multiplexing with guaranteed availability of network resources?

(II) If there are 20 users sharing the link simultaneously, approximately what percentage of time the link will see more transmitting users than it can sustain?

(4b) Consider Node A wants to send a frame carrying data D= 101111000 to Node B using the CRC generator string G = 10111. Node A would like to add some Cyclic Redundancy Check (redundant) bits. What is the maximum number of CRC (redundant) bits to be used?

(4c) An organization is assigned a block of addresses which includes 198.100.101.100/22 as one of the IP addresses. You are required to write the broadcast address for this block.

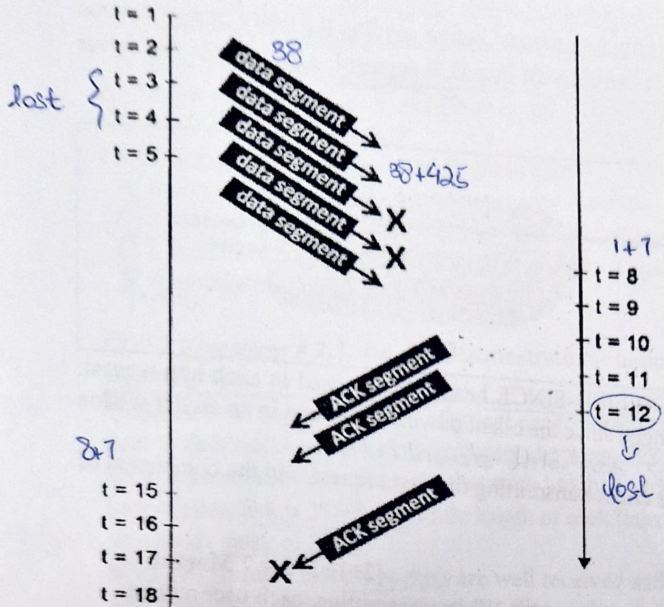
(4d) Suppose host **A** with MAC address **00:1A:2B:3C:4D:5E** on a subnet has IPv4 address as 192.168.0.10/24. Host **B** on the same subnet wants to send a packet to Host A but has no entry for this IP address in its table. Describe the process host B uses for successful communication with host A. Avoid unnecessary details.

Question # 5: Answer the following questions: **[4 + 8 = 12 Marks]**

(5a) A sender is transmitting data to a receiver using sliding window protocols. Window size (N) is 5. The total number of packets to be sent are 15 (numbered as 1 to 15). Assume that if the packet is not lost, TCP receiver immediately sends its acknowledgement. **[2 + 2 = 4]**

- **(i)** If selective repeat is used and packet numbered 5 and 9 are lost. How many packets will the sender retransmit in total and which ones?
- **(ii)** If Go-back-N is used and packet 6 is lost. How many packets will the sender retransmit and which ones?

(5b) As shown in the diagram, a TCP sender has sent 5 segments to the receiver at t = 1, 2, 3, 4, and 5, respectively. Suppose the initial value of the sequence number is 38 and every segment sent to the receiver contains 425 bytes. The delay between the sender and receiver is 7 units, and so the first segment arrives at the receiver at t = 8, and an ACK for this segment arrives at t = 15. Segments sent at t = 3 and t=4 are lost between the sender and the receiver, and one of the ACKs (sent at t=12) gets lost. Assume that there are no timeouts and any out of order segments received are thrown out.



[4+2+2 = 8]

(i) What are the sequence numbers of the packets sent by the sender at t = 1, 2, 3, 4.

(ii) What are the Acknowledgment numbers sent by the receiver at t = 8, 12.

(iii) What will be the sequence numbers at t= 15 and 16.

CLO 3 (Question # 6,7 & 8): **Demonstrate** various classical routing and switching protocols via simulations.

Question # 6: Answer the following questions: **[5+ 7 = 12 Marks]**

(6a): Suppose that only three entries exist in the router's forwarding table as shown below.

Router Forwarding Table

Address	Router Port
129.59.28.0/22	Port 0
129.59.28.0/23	Port 1
129.59.28.0/24	Port 2

The router receives five packets with different destination IP addresses (as provided in I to V). For each packet with a given destination IP address, you are required to consult the router's forwarding table, identify and write the router interface port to which the packets will be forwarded. [1 x 5 = 5]

- **I.** Packet 1: Destination IP = 129.59.29.108. Which router port is this packet forwarded to?
- **II.** Packet 2: Destination IP = 129.59.24.110. Which router port is this packet forwarded to?
- **III.** Packet 3: Destination IP = 129.59.30.57. Which router port is this packet forwarded to?
- **IV.** Packet 4: Destination IP = 129.59.28.250. Which router port is this packet forwarded to?
- **V.** Packet 5: Destination IP = 129.59.28.256. Which router port is this packet forwarded to?

Class B

(6b) A host wants to send a 5020-byte datagram (total datagram length with no options used) over an IPv4 network with a maximum transmission unit (MTU) of 2000 bytes. Thus, the datagram needs to be fragmented. Answer the following questions for the fragmentation process: **[1 + 3 + 3 = 7]**

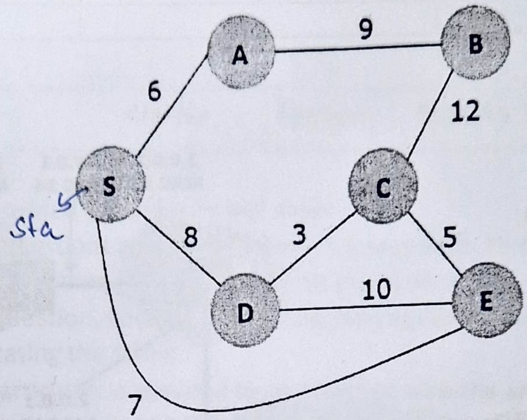
(i) How many total fragments will be made?

(ii) What is the total size (in bytes) of each fragment respectively?

(iii) What is the offset value in each fragment respectively?

Question # 7: Answer the following questions: **[8+1+1 = 10 Marks]**

(7a) Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from S to all network nodes. Provide all steps in a table and final diagram that shows the shortest path from S to all network nodes. **[5+3=8]**

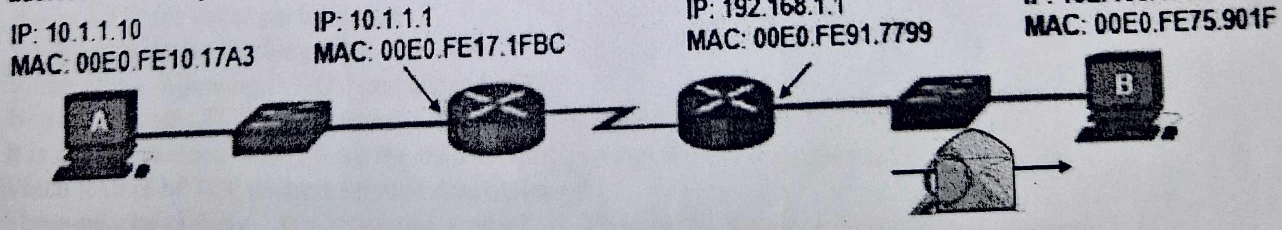


(7b) Which of the OSPF and BGP protocols is based on link state routing algorithm? **[1]**

(7c) Which of the intra-AS and inter-AS routing is policy oriented? **[1]**

Question # 8: Answer the following questions: **[2 + 9 = 11 Marks]**

(8a) Refer to the diagram below. Host A has sent a packet to host B. What will be the source MAC and IP addresses on the packet when it arrives at host B?



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(8b) The figure below shows a network topology. The LAN on the left uses a NAT to connect to the Internet and includes a client host H1. The LAN on the right includes a webserver H2. Packets between the two endpoints are routed along the path shown by heavy dark lines. The various network interfaces have IP and MAC addresses as shown in the figure below.

H1 has established an HTTP session with web server H2 and data packets are flowing between the two machines.

You are required to fill in the header type and the source and destination address for the datalink and network layer headers for packets 1, 2, and 4 (these packets are traveling either from client H1 to server H2 or from server H2 to client H1, as marked on the figure with heavy black arrows and numbers). Note that you should order your headers, i.e., Ethernet should be listed before IP, because the Ethernet packet exists first on the wire.

Write your answers for each packet as per the following template.

Header Type	Source Address	Destination Address